

Preliminary OpenMC Scoping

Here, we use a low-resolution OpenMC model of Excalibur to get first-order estimates of uranium reaction rates and thus scope the quantity of materials necessary.

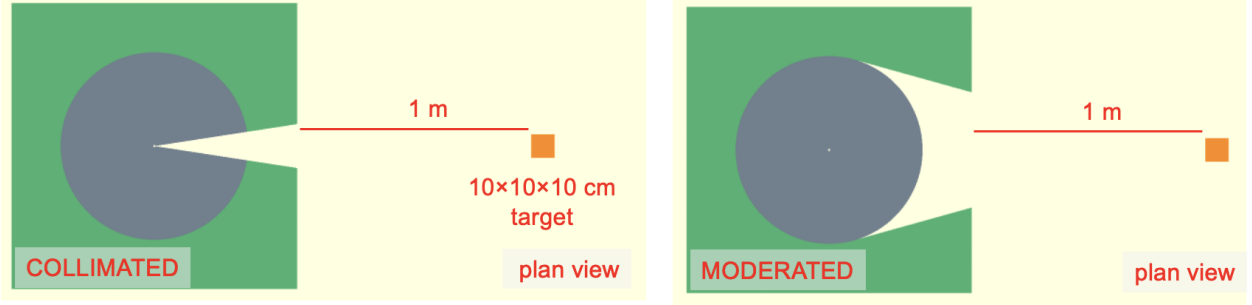


Table 1: Uranium reaction rates are generally low.

Mode	Target depth	UX mol%	Σ_γ [1/cm]			Σ_{fis} [1/cm]		
			CLiF	FLiBe	Factor	CLiF	FLiBe	Factor
Col.	10 cm	0.5	$3.600 \cdot 10^{-6}$	$4.115 \cdot 10^{-6}$	0.9×	$1.563 \cdot 10^{-4}$	$5.856 \cdot 10^{-5}$	2.7×
		5.0	$3.622 \cdot 10^{-5}$	$2.749 \cdot 10^{-5}$	1.3×	$1.365 \cdot 10^{-3}$	$5.737 \cdot 10^{-4}$	2.4×
		10.0	$7.041 \cdot 10^{-5}$	$4.919 \cdot 10^{-5}$	1.4×	$2.392 \cdot 10^{-3}$	$1.128 \cdot 10^{-3}$	2.1×
	20 cm	0.5	$3.819 \cdot 10^{-6}$	$3.634 \cdot 10^{-6}$	1.1×	$1.523 \cdot 10^{-4}$	$5.685 \cdot 10^{-5}$	2.7×
		5.0	$3.811 \cdot 10^{-5}$	$2.712 \cdot 10^{-5}$	1.4×	$1.331 \cdot 10^{-3}$	$5.562 \cdot 10^{-4}$	2.4×
		10.0	$7.642 \cdot 10^{-5}$	$5.036 \cdot 10^{-5}$	1.5×	$2.320 \cdot 10^{-3}$	$1.093 \cdot 10^{-3}$	2.1×
	30 cm	0.5	$3.817 \cdot 10^{-6}$	$3.515 \cdot 10^{-6}$	1.1×	$1.514 \cdot 10^{-4}$	$5.672 \cdot 10^{-5}$	2.7×
		5.0	$3.803 \cdot 10^{-5}$	$2.648 \cdot 10^{-5}$	1.4×	$1.324 \cdot 10^{-3}$	$5.553 \cdot 10^{-4}$	2.4×
		10.0	$7.711 \cdot 10^{-5}$	$5.079 \cdot 10^{-5}$	1.5×	$2.303 \cdot 10^{-3}$	$1.085 \cdot 10^{-3}$	2.1×
Mod.	10 cm	0.5	$6.778 \cdot 10^{-5}$	$2.793 \cdot 10^{-5}$	2.4×	$1.046 \cdot 10^{-5}$	$4.097 \cdot 10^{-6}$	2.6×
		5.0	$6.038 \cdot 10^{-4}$	$2.794 \cdot 10^{-4}$	2.2×	$9.670 \cdot 10^{-5}$	$4.014 \cdot 10^{-5}$	2.4×
		10.0	$9.620 \cdot 10^{-4}$	$5.319 \cdot 10^{-4}$	1.8×	$1.744 \cdot 10^{-4}$	$7.965 \cdot 10^{-5}$	2.2×
	20 cm	0.5	$6.124 \cdot 10^{-5}$	$2.093 \cdot 10^{-4}$	0.3×	$9.799 \cdot 10^{-6}$	$4.062 \cdot 10^{-6}$	2.4×
		5.0	$6.221 \cdot 10^{-4}$	$2.846 \cdot 10^{-4}$	2.2×	$9.049 \cdot 10^{-5}$	$4.047 \cdot 10^{-5}$	2.2×
		10.0	$9.888 \cdot 10^{-4}$	$5.414 \cdot 10^{-4}$	1.8×	$1.704 \cdot 10^{-4}$	$7.992 \cdot 10^{-5}$	2.1×
	30 cm	0.5	$6.636 \cdot 10^{-5}$	$1.921 \cdot 10^{-4}$	0.3×	$1.001 \cdot 10^{-5}$	$4.066 \cdot 10^{-6}$	2.5×
		5.0	$5.958 \cdot 10^{-4}$	$2.718 \cdot 10^{-4}$	2.2×	$8.794 \cdot 10^{-5}$	$4.068 \cdot 10^{-5}$	2.2×
		10.0	$9.516 \cdot 10^{-4}$	$5.247 \cdot 10^{-4}$	1.8×	$1.648 \cdot 10^{-4}$	$8.011 \cdot 10^{-5}$	2.1×